$$2\cot\theta = \frac{P^2-1}{P} \qquad \ldots(4)$$

Dividing (4) by (3) we get, $\dfrac{2\cot\theta}{2\,\text{cosec}\,\theta} = \dfrac{P^2-1}{P} \times \dfrac{P}{P^2+1} \Rightarrow \cos\theta = \dfrac{P^2-1}{P^2+1}$

Example 6.12 Prove that $\tan^2 A - \tan^2 B = \dfrac{\sin^2 A - \sin^2 B}{\cos^2 A \cos^2 B}$

Solution $\tan^2 A - \tan^2 B = \dfrac{\sin^2 A}{\cos^2 A} - \dfrac{\sin^2 B}{\cos^2 B}$

$$= \frac{\sin^2 A\cos^2 B - \sin^2 B\cos^2 A}{\cos^2 A\cos^2 B}$$

$$= \frac{\sin^2 A(1-\sin^2 B) - \sin^2 B(1-\sin^2 A)}{\cos^2 A\ \cos^2 B}$$

$$= \frac{\sin^2 A - \sin^2 A\sin^2 B - \sin^2 B + \sin^2 A\sin^2 B}{\cos^2 A\cos^2 B} = \frac{\sin^2 A - \sin^2 B}{\cos^2 A\cos^2 B}$$

Example 6.13 Prove that $\left(\dfrac{\cos^3 A - \sin^3 A}{\cos A - \sin A}\right) - \left(\dfrac{\cos^3 A + \sin^3 A}{\cos A + \sin A}\right) = 2\sin A\cos A$

Solution $\left(\dfrac{\cos^3 A - \sin^3 A}{\cos A - \sin A}\right) - \left(\dfrac{\cos^3 A + \sin^3 A}{\cos A + \sin A}\right)$

$$= \left(\frac{(\cos A - \sin A)(\cos^2 A + \sin^2 A + \cos A\sin A)}{\cos A - \sin A}\right) \quad \left[\because a^3 - b^3 = (a-b)(a^2+b^2+ab),\ a^3+b^3 = (a+b)(a^2+b^2-ab)\right]$$

$$- \left(\frac{(\cos A + \sin A)(\cos^2 A + \sin^2 A - \cos A\sin A)}{\cos A + \sin A}\right)$$

$$= (1 + \cos A\sin A) - (1 - \cos A\sin A)$$

$$= 2\ \cos A\sin A$$

Example 6.14 Prove that $\dfrac{\sin A}{\sec A + \tan A - 1} + \dfrac{\cos A}{\text{cosec A} + \cot A - 1} = 1$

Solution $\dfrac{\sin A}{\sec A + \tan A - 1} + \dfrac{\cos A}{\text{cosec A} + \cot A - 1}$

$$= \frac{\sin A(\text{cosec}\,A + \cot A - 1) + \cos A(\sec A + \tan A - 1)}{(\sec A + \tan A - 1)(\text{cosec}\,A + \cot A - 1)}$$

$$= \frac{\sin A\,\text{cosec}\,A + \sin A\cot A - \sin A + \cos A\sec A + \cos A\tan A - \cos A}{(\sec A + \tan A - 1)(\text{cosec}\,A + \cot A - 1)}$$

$$= \frac{1 + \cos A - \sin A + 1 + \sin A - \cos A}{\left(\dfrac{1}{\cos A} + \dfrac{\sin A}{\cos A} - 1\right)\left(\dfrac{1}{\sin A} + \dfrac{\cos A}{\sin A} - 1\right)}$$

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